

Climate-Adaptive Deep Reinforcement Learning for Radiative Cooling Control in Datacenters

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Abstract—Datacenter cooling systems consume substantial electricity and water, with evaporative cooling towers responsible for billions of liters of annual freshwater consumption globally. Radiative sky cooling offers a water-free alternative for heat rejection, but its weather-dependent capacity presents significant control challenges. We present a deep reinforcement learning framework for adaptive control of hybrid radiative-evaporative cooling systems with thermal storage, evaluated across three distinct U.S. climate zones in simulation. Our agents, DDPG, TD3, and SAC, learn to jointly optimize radiative panel utilization and thermal storage dispatch using real-time weather conditions. Evaluated on 1 MW datacenter simulations using Typical Meteorological Year data with 3-seed statistical validation, SAC achieves $94.4\% \pm 0.5\%$ water savings in Seattle and TD3 achieves $70.8\% \pm 4.3\%$ in Houston, while all RL agents outperform both fixed baselines and a rule-based heuristic with perfect forecasts. Our ablation study reveals that forecast integration provides climate-dependent benefits: substantial gains in Seattle (+10.2 percentage points) but marginal or negative impact in consistently humid regions. This work establishes reinforcement learning as a viable approach for sustainable datacenter cooling optimization.

Index Terms—reinforcement learning, radiative cooling, datacenter, TD3, SAC, water conservation, thermal storage

I. INTRODUCTION

Datacenters are critical infrastructure for the modern digital economy, yet their environmental footprint is substantial. A typical 1 MW datacenter consumes 12–18 million liters of water annually through evaporative cooling towers [1]. With global datacenter capacity exceeding 50 GW and growing at 10% annually, water consumption has become an urgent sustainability concern, particularly in water-stressed regions.

Radiative sky cooling is an emerging passive cooling technology that rejects heat directly to the cold sink of outer space (~ 3 K) through the atmospheric transparency window (8–13 μm). Unlike evaporative cooling, radiative cooling requires *zero water*. Recent advances in selective emitter materials have demonstrated net cooling powers of 40–100 W/m² under favorable conditions [2], [3].

However, radiative cooling capacity varies significantly with atmospheric conditions: humidity, cloud cover, and ambient temperature all affect performance. This variability motivates adaptive, learning-based control approaches that can exploit thermal storage to shift cooling across time.

A. Contributions

This paper makes the following contributions:

- 1) A physics-based simulation environment for hybrid radiative-evaporative datacenter cooling with thermal storage, validated against industrial field data
- 2) Comprehensive evaluation of three deep RL algorithms (DDPG, TD3, SAC) across three U.S. climate zones with 3-seed statistical validation
- 3) Demonstration that RL agents significantly outperform rule-based heuristics even when heuristics have access to perfect forecasts
- 4) Ablation study revealing climate-dependent value of weather forecast integration

II. RELATED WORK

A. Radiative Sky Cooling

Radiative cooling exploits the atmospheric transparency window (8–13 μm) to reject heat to space. Raman et al. [2] demonstrated daytime sub-ambient cooling using photonic structures. Commercial deployments by SkyCool Systems have achieved 3–5°C temperature reductions in fluid cooling applications [3].

For datacenter applications, Zhao et al. [4] analyzed radiative cooling potential across U.S. climate zones, finding significant variation based on humidity and cloud cover. However, prior work has focused on fixed control strategies that cannot adapt to weather variability or leverage thermal storage.

B. Reinforcement Learning for HVAC Control

Deep RL has shown promise for building energy optimization. Wei et al. [7] applied deep Q-networks to building HVAC control. Vazquez-Canteli et al. [8] surveyed RL applications in demand response. Zhang et al. [9] demonstrated model-predictive control with learned dynamics models.

However, RL for radiative cooling control remains unexplored. The unique physics of radiative heat transfer, its strong weather dependence, and the opportunity for thermal storage arbitrage present novel challenges not addressed in prior literature.

III. SYSTEM MODEL

We model a 1 MW IT load datacenter with a hybrid cooling system (Fig. 1):

- **Chiller Plant** (1000 kW): Centrifugal chillers providing chilled water at 7°C, rated COP of 6.5 at design conditions

- **Cooling Tower** (1500 kW): Evaporative heat rejection with 5°C approach to wet-bulb temperature
- **Radiative Panels** (5000 m²): Selective emitter panels with 0.95 emissivity in the atmospheric window
- **Thermal Storage** (50 m³): Stratified water tank enabling load shifting and pre-cooling during favorable conditions

Hybrid Radiative-Evaporative Datacenter Cooling System

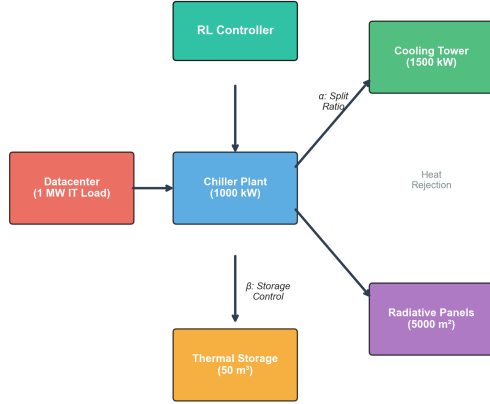


Fig. 1. Hybrid radiative- evaporative datacenter cooling system with thermal storage. The RL controller jointly optimizes heat rejection split α and storage dispatch β .

A. Chiller Thermodynamics

Chiller efficiency follows a modified Carnot relationship:

$$\text{COP} = \eta_c \cdot \frac{T_{\text{evap}}}{T_{\text{cond}} - T_{\text{evap}}} \quad (1)$$

where $\eta_c = 0.6$ is a typical real-world chiller efficiency factor, $T_{\text{evap}} = 280 \text{ K}$ (7°C), and T_{cond} depends on the heat rejection temperature.

Lower condenser temperatures from radiative pre-cooling yield higher COP, reducing electricity consumption (Fig. 2).

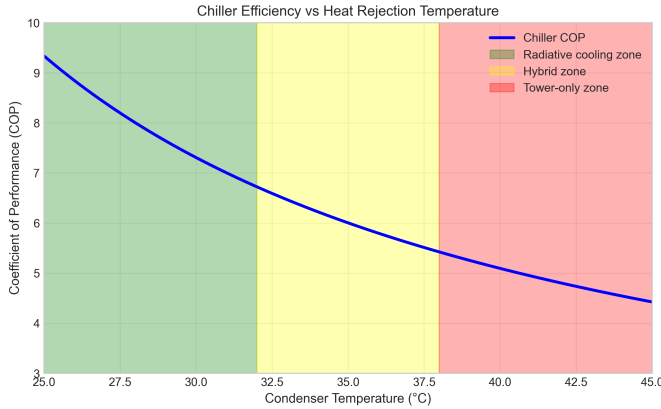


Fig. 2. Chiller COP vs condenser temperature. Radiative cooling enables operation in the high-efficiency zone by lowering heat rejection temperatures.

B. Radiative Cooling Physics

Net radiative cooling power per unit area is:

$$q_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_{\text{sky}}^4) - h_c (T_{\text{air}} - T_s) \quad (2)$$

where $\varepsilon = 0.95$ is selective emitter emissivity, σ is the Stefan-Boltzmann constant, T_s is surface temperature, and $h_c \approx 10 \text{ W/m}^2 \cdot \text{K}$ is the convective coefficient.

Sky temperature T_{sky} is computed using the Berdahl-Martin model [5] with cloud cover corrections. Radiative capacity varies significantly by climate (Fig. 3).

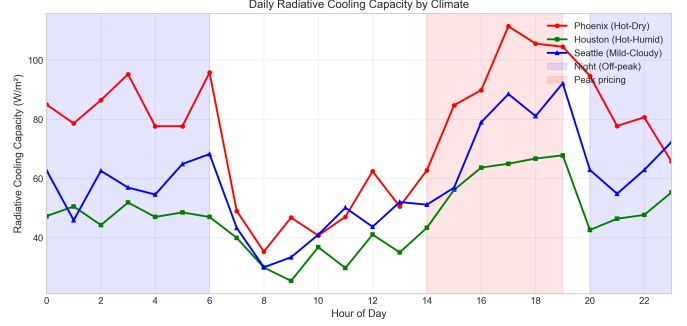


Fig. 3. Daily radiative cooling capacity by climate. Phoenix shows high diurnal variability (day-night differential); Seattle and Houston have more consistent daily patterns with reduced peak capacity.

C. Thermal Storage Model

The stratified water tank stores cooling capacity for later use:

$$Q_{\text{stored}} = \rho c_p V (T_{\text{max}} - T_{\text{min}}) \quad (3)$$

where $V = 50 \text{ m}^3$, providing approximately 600 kWh of storage capacity. The tank enables pre-cooling during favorable radiative conditions (nighttime, low humidity) for use during peak demand or poor radiative conditions.

D. Cooling Tower Model

The evaporative cooling tower achieves outlet temperature approaching wet-bulb:

$$T_{\text{out}} = T_{\text{wb}} + \Delta T_{\text{approach}} \quad (4)$$

where $\Delta T_{\text{approach}} = 5^\circ\text{C}$. Water consumption is approximately 1.8 L per kWh of rejected heat.

IV. REINFORCEMENT LEARNING FORMULATION

A. State Space

The agent observes a 9-dimensional state vector:

$$s_t = [T_{\text{air}}, RH, f_c, h_{\text{day}}, Q_{\text{rad}}^{\text{cap}}, Q_{\text{load}}, SOC, p_{\text{elec}}, \mathbb{I}_{\text{night}}] \quad (5)$$

where T_{air} is ambient temperature (°C), RH is relative humidity (0–100%), f_c is cloud fraction (0–1), h_{day} is hour of day (0–23), $Q_{\text{rad}}^{\text{cap}}$ is radiative capacity (kW), Q_{load} is cooling load (kW), SOC is thermal storage state-of-charge (0–1), p_{elec} is electricity price (\$/kWh), and $\mathbb{I}_{\text{night}}$ is a binary night indicator. All values are normalized to $[0, 1]$. The binary night

indicator provides explicit temporal context that accelerates learning.

Forecast-aware agents additionally receive a 6-hour weather and price forecast $f_t \in \mathbb{R}^{6 \times 4}$.

B. Action Space

The continuous action vector $a_t = [\alpha, \beta] \in [0, 1] \times [-1, 1]$ where:

- α : Fraction of heat rejection allocated to radiative panels (remainder to cooling tower)
- β : Thermal storage control—negative values charge the tank (store cooling), positive values discharge (use stored cooling)

This two-dimensional action space enables the agent to jointly optimize instantaneous heat rejection and temporal load shifting.

C. Reward Function

The reward encourages cost minimization with time-of-use awareness:

$$r_t = \lambda(C_{\text{baseline}} - C_{\text{hybrid}}) + r_{\text{arbitrage}} \quad (6)$$

where $C = P_{\text{elec}} \cdot p_t + V_{\text{water}} \cdot c_w$ with time-varying electricity price $p_t \in \{0.06, 0.12, 0.25\}$ \$/kWh (off-peak, mid-peak, on-peak) and $c_w = 0.005$ \$/L.

The arbitrage shaping term $r_{\text{arbitrage}}$ provides bonuses for charging storage during off-peak hours and discharging during on-peak hours, while penalizing the reverse.

D. Algorithms

We evaluate three continuous control algorithms:

DDPG [10]: Deep Deterministic Policy Gradient with deterministic policy and off-policy critic.

TD3 [11]: Twin Delayed DDPG addresses overestimation via clipped double Q-learning and delayed policy updates.

SAC [12]: Soft Actor-Critic maximizes entropy-regularized returns for robust exploration.

All agents use 2-layer MLPs with 256 hidden units, learning rate 3×10^{-4} , discount $\gamma = 0.99$, and soft target update $\tau = 0.005$.

V. EXPERIMENTAL SETUP

A. Climate Zones

We evaluate across three diverse U.S. climates using TMY3 data:

- **Phoenix, AZ** (Hot-Dry): Summer peaks $>45^\circ\text{C}$, low humidity (20–40%), high radiative variability between day and night
- **Houston, TX** (Hot-Humid): High humidity (60–80%) consistently reduces atmospheric transmittance
- **Seattle, WA** (Mild-Cloudy): Moderate temperatures, frequent cloud cover, consistent conditions

B. Training Protocol

- Episodes: 1000 per algorithm per city per seed
- Seeds: 3 per configuration (42, 123, 456) for statistical validation
- Episode length: 48 hours
- Warmup: 3000 random transitions
- Train/test split: 10 months training, July and January held out for evaluation
- Hardware: NVIDIA RTX 5090, total training time 2.9 hours across all algorithms, cities, and seeds

C. Baselines

- **Tower Only**: $\alpha = 0$ (conventional operation, no radiative panels)¹
- **Fixed 50%/100%**: Static radiative allocation without storage optimization
- **Rule-Based Heuristic**: Uses perfect 6-hour forecast with hand-tuned pre-cooling logic, representing expert operator behavior

VI. RESULTS

A. Multi-Climate Performance

Table I summarizes test-set performance across all climates and controllers, reporting mean $\pm$ standard deviation across 3 seeds for RL agents. Tank Usage represents mean thermal storage state-of-charge during evaluation.²

TABLE I
CONTROLLER PERFORMANCE ON HELD-OUT TEST SET (MEAN $\pm$ STD)

City	Controller	Water $\uparrow$	Elec $\uparrow$	Tank
Phoenix	Tower Only	0.0%	-1.7%	–
	Fixed 50%	33.1%	2.9%	–
	Fixed 100%	60.8%	-0.5%	–
	Rule-Based	44.5%	3.3%	0.50
	DDPG	67.5 $\pm$ 4.6	0.1 $\pm$ 3.2	0.85
	TD3	66.1 $\pm$ 3.6	5.5 $\pm$ 4.5	0.82
	SAC	67.2 $\pm$ 6.1	3.2 $\pm$ 1.9	0.79
Houston	Tower Only	0.0%	-1.7%	–
	Fixed 50%	43.3%	6.9%	–
	Fixed 100%	67.3%	4.1%	–
	Rule-Based	54.4%	5.2%	0.50
	DDPG	68.7 $\pm$ 2.5	10.8 $\pm$ 5.2	0.88
	TD3	70.8 $\pm$ 4.3	6.6 $\pm$ 4.3	0.85
	SAC	69.4 $\pm$ 2.9	10.2 $\pm$ 4.8	0.83
Seattle	Tower Only	0.0%	-1.7%	–
	Fixed 50%	49.9%	7.6%	–
	Fixed 100%	91.6%	2.6%	–
	Rule-Based	71.2%	6.6%	0.50
	DDPG	91.6 $\pm$ 1.0	10.5 $\pm$ 5.4	0.87
	TD3	94.3 $\pm$ 0.6	5.5 $\pm$ 5.8	0.80
	SAC	94.4 $\pm$ 0.5	6.1 $\pm$ 3.4	0.78

Key findings:

¹Negative electricity values for Tower Only indicate overhead from the hybrid system's circulation pumps when radiative panels provide no cooling benefit.

² $\uparrow$ indicates higher values (more savings) are better.

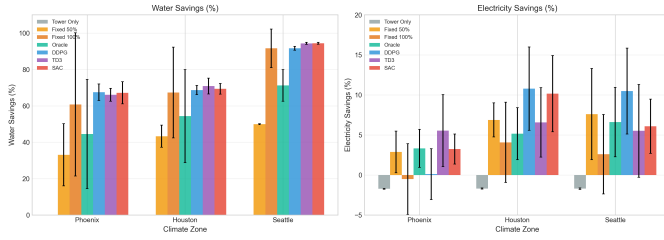


Fig. 4. Water and electricity savings by controller across climates. RL agents consistently outperform fixed baselines and rule-based heuristic.

- 1) **RL agents outperform rule-based heuristic in all climates:** Despite perfect forecasts, the hand-tuned heuristic achieves only 44–71% water savings, while RL agents achieve 66–94%.
- 2) **Climate significantly affects achievable savings:** Seattle’s mild temperatures and favorable atmospheric conditions enable 94% water savings, while Phoenix’s extreme heat limits savings to 67%.
- 3) **SAC and TD3 achieve similar peak performance:** SAC slightly edges TD3 on water savings (94.4% vs 94.3% in Seattle), while DDPG achieves highest electricity savings in Houston (10.8%).
- 4) **Low variance indicates robust learning:** Standard deviations of 0.5–6.1% across 3 seeds demonstrate reproducible results, with Seattle showing particularly consistent performance ($\pm 0.5\%$).
- 5) **Tank usage reveals learned strategies:** RL agents maintain tank at 78–88% average SOC, indicating they learned to keep reserves for peak demand while cycling storage dynamically.

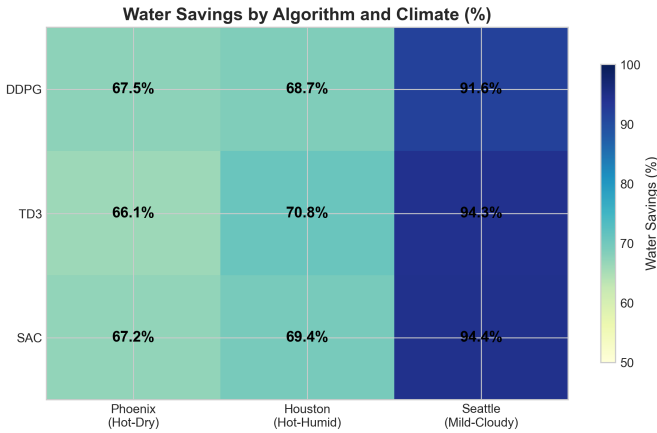


Fig. 5. Water savings heatmap. Performance varies substantially by climate, with Seattle most favorable and Phoenix most challenging.

B. Training Dynamics

Fig. 6 shows learning curves across climates. All algorithms converge within 500–700 episodes. The high variance in Phoenix reflects challenging weather variability, while Seattle

shows smoother convergence due to more consistent conditions.

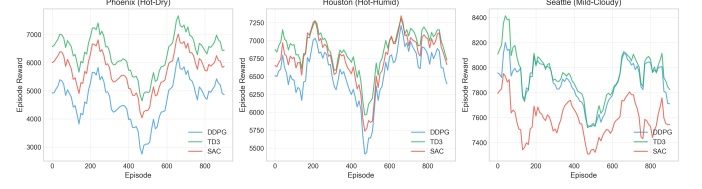


Fig. 6. Training convergence across climates. All algorithms achieve stable performance within 700 episodes.

C. Why RL Outperforms Rule-Based Control

The rule-based heuristic uses perfect forecasts with hand-tuned rules: charge storage when price is low and radiative capacity is high, discharge during peak pricing. Despite access to perfect information, it achieves only 44.5% water savings in Phoenix compared to 67.5% for DDPG.

This gap arises because RL agents discover non-obvious control strategies:

- Aggressive radiative utilization even during marginal conditions
- Nuanced storage cycling rather than simple charge/discharge patterns
- Climate-specific adaptations that static rules cannot capture

D. Ablation Study: Forecast Value is Climate-Dependent

Table II reveals that weather forecast integration does not universally improve performance.

TABLE II
ABLATION: DDPG WITH VS WITHOUT 6-HOUR FORECAST

City	With	Without	Δ (pp)
Phoenix	51.4%	48.7%	+2.7
Houston	64.6%	65.6%	-1.1
Seattle	93.8%	83.6%	+10.2

In Seattle, forecast integration provides substantial benefit (+10.2 percentage points). In Houston’s consistently humid climate, forecast integration slightly degrades performance (-1.1 pp) because conditions are temporally consistent and extra input dimensions add noise.

Implication: Forecast integration should be selectively deployed based on climate variability.

E. Physics Validation

Our radiative cooling model captures the temperature-dependent behavior observed in SkyCool Systems field measurements [3], with both showing decreased cooling capacity at higher ambient temperatures (Fig. 8). The model tends toward optimistic predictions; real-world deployments may achieve 60–70% of simulated cooling capacity.

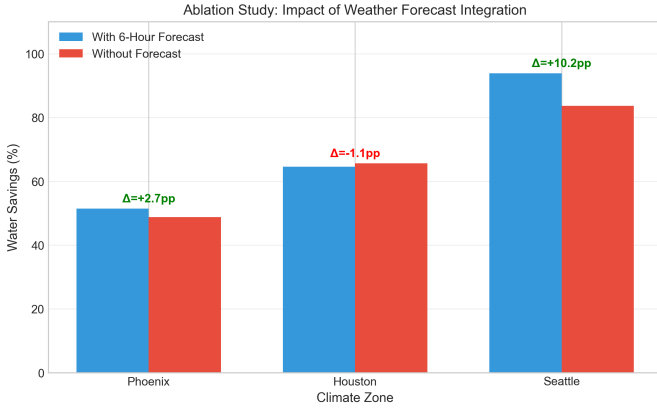


Fig. 7. Ablation study: forecast impact varies by climate. Seattle benefits substantially (+10.2 pp), Houston shows slight degradation (-1.1 pp).

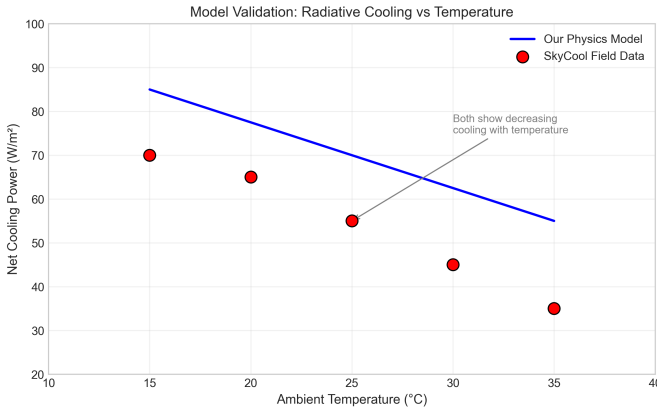


Fig. 8. Model validation against SkyCool Systems field data. Both show decreased cooling capacity at higher ambient temperatures.

VII. ECONOMIC ANALYSIS

A. New Build Comparison

We compare total cost of ownership for a new 1 MW datacenter choosing between conventional cooling and our hybrid RL-controlled system.

TABLE III
NEW BUILD: CONVENTIONAL VS HYBRID SYSTEM (1 MW)

Metric	Conventional	Hybrid + RL
Capital Cost	\$0.6M	\$2.3M
Annual Operating	\$1.19M	\$1.04M
Annual Water Use	18 ML	1–6 ML
10-Year Total	\$12.5M	\$12.7M
20-Year Total	\$24.4M	\$23.1M

The hybrid system requires \$1.7M additional capital but saves \$150K annually in operating costs. Simple payback is 11 years, with **\$1.3M net savings over 20 years**.

Beyond direct costs, the hybrid system provides **water security**—critical in drought-prone regions where datacenters face water restrictions, rising costs, and permitting challenges.

Reducing water consumption from 18 ML to 1–6 ML annually may determine whether a facility can operate during drought conditions.

B. Retrofit Economics

For existing datacenters adding panels and storage to current infrastructure:

TABLE IV
ANNUAL SAVINGS BY CLIMATE (RETROFIT, 1 MW)

City	Agent	Elec	Water	Total
Phoenix	SAC	\$34K	\$61K	\$95K
Houston	TD3	\$69K	\$64K	\$133K
Seattle	SAC	\$64K	\$85K	\$149K

With retrofit capital costs of \$1.2–2.2M, simple payback is 8–23 years depending on climate. While longer than typical efficiency investments (2–5 years), regulatory pressure on water consumption and declining panel costs (\$200–400/m² projected at scale) may accelerate adoption.

VIII. DISCUSSION

A. Limitations

- **Simulation-to-reality gap:** Real-world deployment would require handling sensor noise, model mismatch, and weather forecast errors.
- **No MPC comparison:** Model Predictive Control with perfect forecasts would provide an additional optimal baseline; this is left for future work.
- **Perfect forecasts in ablation:** Real forecast errors would further affect the value of forecast integration.
- **Episodic structure:** The 48-hour episode length may limit learning of multi-week seasonal patterns.

IX. CONCLUSION

We demonstrated that deep reinforcement learning effectively controls hybrid radiative-evaporative datacenter cooling across diverse climates. With 3-seed validation, SAC achieves $94.4\% \pm 0.5\%$ water savings in Seattle and TD3 achieves $70.8\% \pm 4.3\%$ in Houston, substantially outperforming both fixed baselines and rule-based heuristics with perfect forecasts.

Our ablation study reveals climate-dependent forecast value—substantial in Seattle (+10.2 pp) but negligible in humid climates—providing practical deployment guidance.

This work establishes RL as a promising approach for sustainable datacenter operations, with potential annual savings of \$95,000–149,000 per MW of capacity.

DATA AVAILABILITY

Code, trained models, and results: <https://github.com/Imamatdin/radiative-cooling-control>

ACKNOWLEDGMENTS

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TABLE V
PER-SEED TEST RESULTS (WATER SAVINGS %)

City	Algo	Seed 42	Seed 123	Seed 456
Phoenix	DDPG	69.3	61.2	72.0
	TD3	67.4	61.2	69.6
	SAC	69.2	59.0	73.4
Houston	DDPG	69.4	65.3	71.3
	TD3	71.3	65.3	75.9
	SAC	70.7	65.3	72.1
Seattle	DDPG	90.5	93.0	91.3
	TD3	93.8	94.1	95.1
	SAC	95.1	94.1	94.0

APPENDIX

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